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## Quantum Physics Pset 1

1 (a)  $z_1 = 4e^{i\pi/3}$ ,  $z_2 = -3 + 4i$ ,  $z_3 = \frac{1}{1-2i}$ ,  $z_4 = (3-2i)^2$

(i) real & imaginary parts?

$$z_1 = 2 + i \times 2\sqrt{3} \Rightarrow \text{real: } 2, \text{ imaginary: } 2\sqrt{3}$$

$$z_2 = -3 + 4i \Rightarrow \text{real: } -3, \text{ imaginary: } 4$$

$$z_3 = \frac{1+2i}{1+4} = \frac{1}{5} + \frac{2}{5}i \Rightarrow \text{real: } \frac{1}{5}, \text{ imaginary: } \frac{2}{5}$$

$$z_4 = 9 + 4 \times (-1) - 12i = 5 - 12i \Rightarrow \text{real: } 5, \text{ imaginary: } -12$$

(ii)  $z^*$ ?

$$z_1^* = 2 - 2\sqrt{3}i$$

$$z_2^* = -3 - 4i$$

$$z_3^* = \frac{1}{5} - \frac{2}{5}i$$

$$z_4^* = 5 + 12i$$

(iii)  $|z|^2 \equiv z^* z$ ?

$$|z_1|^2 = 4 + 4 \cdot 3 = 16$$

$$|z_2|^2 = 9 + 16 = 25$$

$$|z_3|^2 = \frac{1}{25} + \frac{4}{25} = \frac{5}{25} = \frac{1}{5}$$

$$|z_4|^2 = 25 + 144 = 169$$

(iv)  $|z|$ ?

$$|z_1| = 4$$

$$|z_2| = 5$$

$$|z_3| = \frac{1}{\sqrt{5}}$$

$$|z_4| = 13$$

(v)  $z_1 = 2 + 2\sqrt{3}i = 4e^{i\pi/3}$

$$z_2 = -3 + 4i = 5e^{2.214i}$$

$$z_3 = \frac{1}{5} + \frac{2}{5}i = \frac{1}{\sqrt{5}}e^{1.107i}$$

$$z_4 = 5 - 12i = 13e^{-1.176i}$$

(b)  $z_1 = e^{-ikx}$ ,  $z_2 = e^{iky}$

$|z_1 + z_2|^2$  ?

$= |z_1|^2 + |z_2|^2 + z_1 z_2^* + z_1^* z_2$

$= 1 + 1 + (\cos(kx) - i\sin(kx))(\cos(ky) - i\sin(ky))$   
 $+ (\cos(kx) + i\sin(kx))(\cos(ky) + i\sin(ky))$

$= 2 + 2\cos(kx)\cos(ky) - 2\sin(kx)\sin(ky)$

$= 2 + 2\cos(k(x+y))$

$|z_1 - z_2|^2$  ?

$= |z_1|^2 + |z_2|^2 - z_1 z_2^* - z_1^* z_2$

$= 2 - 2\cos(k(x+y))$

$z_1 z_2^* - z_1^* z_2$  ?

$= (\cos(kx) - i\sin(kx))(\cos(ky) - i\sin(ky)) -$   
 $(\cos(kx) + i\sin(kx))(\cos(ky) + i\sin(ky))$

$= -2i\sin(ky)\cos(kx) - 2i\sin(kx)\cos(ky)$

$= -2i\sin(k(x+y))$

2. (i)  $5 \cdot 10^{-17} \text{ J}$  to eV

$$\frac{5 \times 10^{-17}}{1.602 \times 10^{-19}} = 312.1 \text{ eV}$$

(ii)  $\lambda = 500 \text{ nm}$

$$\frac{hc}{\lambda} = \frac{1240}{500} = 2.48 \text{ eV}$$

(iii)  $10 \mu\text{W}$  to eV/s

$$\frac{10 \times 10^{-6}}{1.602 \times 10^{-19}} = 6.24 \times 10^{13} \text{ eV/s}$$

(iv)  $\nu = 50 \text{ GHz}$ .  $k$ ?

$$k = \frac{2\pi\nu}{c} = \frac{2\pi \times 50 \times 10^9}{3 \times 10^8} = 1047.2 \text{ rad/m}$$

3.  $\rho(x) = A e^{-\lambda|x-a|}$

(i)  $A$ ?

$$\int_{-\infty}^{\infty} \rho(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} A e^{-\lambda|x-a|} dx = 1 \Rightarrow A \int_{-\infty}^{\infty} e^{-\lambda|x-a|} dx = 1$$

let  $u = x - a$ ,  $du = dx$ .

$$\Rightarrow A \int_{-\infty}^{\infty} e^{-\lambda|u|} du = 1 \Rightarrow 2A \int_0^{\infty} e^{-\lambda u} du = 1 \Rightarrow 2A \left[ \frac{e^{-\lambda u}}{-\lambda} \right]_0^{\infty} = 1$$

$$\Rightarrow 2A \left( 0 - \left( -\frac{1}{\lambda} \right) \right) = \frac{2A}{\lambda} = 1 \Rightarrow A = \boxed{\frac{\lambda}{2}}$$

(ii)  $\langle x \rangle$ : by symmetry,  $\langle x \rangle = \boxed{a}$

$$\langle x^2 \rangle: \int x^2 \rho(x) dx$$

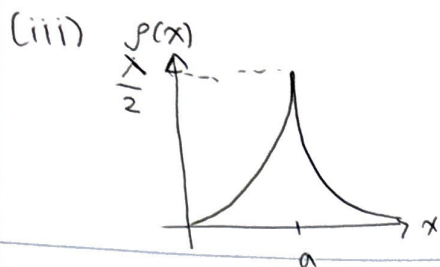
$$= \frac{\lambda}{2} \int_{-\infty}^{\infty} x^2 e^{-\lambda|x-a|} dx$$

let  $u = x - a \Rightarrow x = u + a$

$$\Rightarrow \frac{\lambda}{2} \int_{-\infty}^{\infty} (u+a)^2 e^{-\lambda|u|} du = \frac{\lambda}{2} \left[ \int_{-\infty}^{\infty} u^2 e^{-\lambda|u|} du + \int_{-\infty}^{\infty} 2au e^{-\lambda|u|} du + \int_{-\infty}^{\infty} a^2 e^{-\lambda|u|} du \right]$$

$$= a^2 + \frac{\lambda}{2} \int_{-\infty}^{\infty} u^2 e^{-\lambda|u|} du = a^2 + \lambda \left( \frac{2}{\lambda^3} \right) = \boxed{a^2 + \frac{2}{\lambda^2}}$$

$$\sigma: \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\left( a^2 + \frac{2}{\lambda^2} \right) - a^2} = \sqrt{\frac{2}{\lambda^2}} = \boxed{\frac{\sqrt{2}}{\lambda}}$$



4.  $\Psi(x, t) = \begin{cases} A \cos(\pi x/a) e^{-2i\omega t} & , -a/2 \leq x \leq a/2 \\ 0 & , \text{otherwise} \end{cases}$

(i) normalize

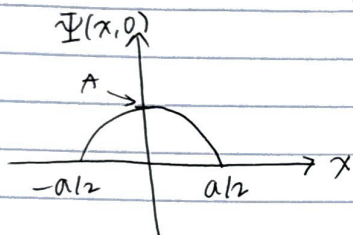
$$A^2 \int_{-a/2}^{a/2} \cos^2(\pi x/a) dx = 1$$

$$\therefore |e^{-2i\omega t}|^2 = 1.$$

$$= \frac{A^2}{2} \int_{-a/2}^{a/2} 1 + \cos\left(\frac{2\pi x}{a}\right) dx = \frac{A^2}{2} \left[ x + \sin\left(\frac{2\pi x}{a}\right) \right]_{-a/2}^{a/2}$$

$$= \frac{A^2}{2} \left( \frac{a}{2} + 0 - \left(-\frac{a}{2} + 0\right) \right) = \frac{A^2 a}{2} = 1 \Rightarrow A = \sqrt{\frac{2}{a}}$$

(ii)



(iii)  $x=0$  (where the graph above is at maximum)

(iv)  $\langle x \rangle = 0$  (symmetry)

$$\langle x^2 \rangle = \int_{-a/2}^{a/2} \Psi^2 x^2 dx = \frac{2}{a} \int_{-a/2}^{a/2} x^2 \cos^2\left(\frac{\pi x}{a}\right) dx$$

$$= \frac{1}{a} \int_{-a/2}^{a/2} x^2 \left( 1 + \cos\left(\frac{2\pi x}{a}\right) \right) dx = \frac{1}{a} \int_{-a/2}^{a/2} x^2 dx + \frac{1}{a} \int_{-a/2}^{a/2} x^2 \cos\left(\frac{2\pi x}{a}\right) dx$$

$$= \frac{1}{a} \left[ \frac{x^3}{3} \right]_{-a/2}^{a/2} + \frac{1}{a} \left[ \frac{2x \cos(2\pi x/a)}{(2\pi/a)^2} \right]_{-a/2}^{a/2} = \boxed{\frac{a^2}{12} - \frac{a^2}{2\pi^2}}$$

$$\sigma = \sqrt{a^2 \left( \frac{1}{12} - \frac{1}{2\pi^2} \right)}$$

$$= \boxed{a \sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}}$$

(v)

$$1 - \int_{-\sigma}^{\sigma} \frac{2}{a} \cos^2\left(\frac{\pi x}{a}\right) dx = 1 - \frac{1}{a} \left[ x + \frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right) \right]_{-\sigma}^{\sigma}$$

$$= 1 - \frac{2\sigma}{a} - \frac{1}{\pi} \sin\left(\frac{2\pi\sigma}{a}\right) = 0.35$$

$$\therefore 35\%$$